

(TR-104) PYTHON WITH AI/ML

Training Day 71 Report

Date: 20 April 2026

Topic Covered: AI Chatbot System Development using FastAPI, LangChain Agents & MongoDB

The seventy-first day of training focused on understanding and developing an intelligent chatbot system using FastAPI, LangChain agents, MongoDB, and Ollama models. The training also included workflow analysis, automation concepts, and system integration understanding.

- Analyzed the requirements and workflow of an intelligent chatbot system for SuperAdmin operations.
- Studied the complete chatbot flow from frontend interaction to backend AI processing.
- Explored agent-based architecture for database querying and automated reporting.
- Understood chatbot integration with existing web applications.
- Learned the use of cron jobs and cloud scheduling for automation tasks.
- Identified chatbot data requirements related to payments, leases, employees, and users.
- Studied notification systems including email, SMS, and in-app alerts.
- Developed a FastAPI backend application for chatbot query handling.
- Integrated MongoDB using pymongo for dynamic data retrieval.
- Created tools for querying payments, employees, and lease-related data.
- Implemented count_documents() logic for accurate database statistics.
- Designed /chat API endpoint for user query processing.
- Integrated LangChain agents with FastAPI backend for intelligent tool-based responses.
- Configured Ollama model for AI-based query handling and response generation.
- Defined structured system prompts to control tool usage and agent behavior.
- Studied the complete architecture of the Trailer Tracking System project.

- Analyzed modules including trailer management, inventory, insurance, and customer management.
- Understood role-based access control and permission management.
- Designed end-to-end workflow for chatbot interaction and response flow.
- Understood integration flow between React frontend, Node.js backend, FastAPI service, and LangChain agent.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/smart_assistant
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/talk-to-your-database-building-a-smart-data-query-assistant-using-llm-tools-150d66fb9c0f>

Overall, the training strengthened understanding of AI chatbot systems, FastAPI backend development, LangChain agents, MongoDB integration, workflow automation, and intelligent tool-based AI architectures.